The Langlands Thread: Riemann, Yang-Mills, and BSD as Instances of a Single Program

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Abstract

Three of the four remaining open Clay Millennium Problems — the Riemann Hypothesis, Yang-Mills existence and mass gap, and Birch-Swinnerton-Dyer — are connected by the Langlands program. The connection operates at two levels. At the *geometric* level, the Langlands correspondence for complex curves (now proved: Ben-Zvi-Chen-Helm-Nadler [1]) identifies the Yang-Mills mass gap, via Kapustin-Witten [2], with an S-duality statement in $\mathcal{N} = 4$ super Yang-Mills. At the *arithmetic* level, BSD for rank ≥ 2 is exactly the arithmetic Langlands functoriality for $U(r) \times U(r+1)$ (arithmetic GGP), and GRH is equivalent to the Ramanujan conjecture for all automorphic forms on $GL(n)$ over $\mathbb{Q}$. The geometric Langlands program being proved does not close the arithmetic problems: the passage from $\mathbb{F}_q$ and $\mathbb{C}$ to $\mathbb{Q}$ requires Frobenius-type control that is unavailable over number fields. Navier-Stokes has no Langlands analog: it is structurally separate.

1 The Langlands Program: Two Modes

Definition 1 (Langlands data)**.** Let G be a reductive group over a field F, with Langlands dual $G^\vee$. The Langlands correspondence (in various forms) asserts a correspondence between:

- *Automorphic side*: representations π of $G(\mathbb{A}_F)$ (adelic points), from $L^2(G(F)\backslash G(\mathbb{A}_F))$.
- *Galois side*: L-parameters $\varphi : \mathrm{Gal}(\bar{F}/F) \to G^\vee(\mathbb{C})$ (continuous homomorphisms).
- *Matching*: $L(s, \pi) = L(s, \varphi)$ (equality of L-functions).

The program operates in two structurally different regimes:

Geometric Langlands. $F = \mathbb{F}_q(C)$ (function field of a curve C over $\mathbb{F}_q$) or $F = \mathbb{C}(C)$ (complex curve). Frobenius is available over $\mathbb{F}_q$; algebraic geometry dominates. In the complex case, the correspondence takes the categorical form:

$$D(\mathrm{Bun}_G(C)) \;\simeq\; \mathrm{QCoh}(\mathrm{LocSys}_{G^\vee}(C)),$$

a derived equivalence of stable ∞-categories.

Arithmetic Langlands. $F = \mathbb{Q}$ or a number field. Frobenius is only available locally (at primes of good reduction). Global control requires entirely new tools; the correspondence is mostly conjectural.

Theorem 2 (Ben-Zvi-Chen-Helm-Nadler [1])**.** *The geometric Langlands conjecture holds for smooth projective curves over algebraically closed fields of characteristic zero:* $D(\mathrm{Bun}_G(C)) \simeq \mathrm{QCoh}(\mathrm{LocSys}_{G^\vee}(C))$ *as stable* ∞*-categories.*

Remark 3. Theorem 2 completes the geometric program. It does not imply arithmetic Langlands: the passage from $\mathbb{C}(C)$ to $\mathbb{Q}$ has no direct route. The geometric proof uses D-modules and perverse sheaves; arithmetic uses Galois cohomology and p-adic Hodge theory. They are different subjects that share a philosophy.

2 Yang-Mills and Geometric Langlands

Theorem 4 (Kapustin-Witten [2])**.** *The geometric Langlands correspondence is equivalent to S-duality of* $\mathcal{N} = 4$ *super Yang-Mills (SYM) theory with gauge group G on a four-manifold* $M_4 = C \times \mathbb{R}^2$ *(for a Riemann surface C). Specifically:*

- *The S-duality symmetry* $\tau \mapsto -1/(n_G\tau)$ *swaps G and* $G^\vee$*.*
- *D-branes in* $\mathcal{N} = 4$ *SYM correspond to objects in* $D(\mathrm{Bun}_G)$ *and* $\mathrm{QCoh}(\mathrm{LocSys}_{G^\vee})$*.*
- *The categorical equivalence of Theorem 2 is the mathematical statement of S-duality.*

Remark 5 ($\mathcal{N} = 4$ SYM is not pure YM)*.* The Clay Millennium Problem concerns *pure* Yang-Mills with gauge group G (SU(2) or SU(3)) in four dimensions, no supersymmetry. $\mathcal{N} = 4$ SYM is a different theory: it is exactly conformally invariant (the β-function vanishes identically), has no mass gap from dimensional transmutation, and has 16 supercharges. The mass gap problem does not arise for $\mathcal{N} = 4$ SYM because the theory is scale-invariant.

The connection between $\mathcal{N} = 4$ SYM and pure YM goes through *supersymmetry breaking*: deforming $\mathcal{N} = 4$ SYM by a mass term for the adjoint scalars breaks $\mathcal{N} = 4 \to \mathcal{N} = 0$ (pure YM) in the infrared. The S-duality of $\mathcal{N} = 4$ may survive this deformation in a modified form, but this is speculative.

Proposition 6 (What geometric Langlands gives YM)**.** *Theorem 2 (geometric Langlands proved) implies:*

1. *S-duality of* $\mathcal{N} = 4$ *SYM is mathematically substantiated.*
2. *The spectrum of* $\mathcal{N} = 4$ *SYM is invariant under* $G \leftrightarrow G^\vee$*.*
3. *For pure YM: if the mass gap* Δ *is invariant under electric-magnetic duality [3], then* $\Delta(G) = \Delta(G^\vee)$*.*

Remark 7. Point (3) is a symmetry statement, not an existence statement. Even if $\Delta(G) = \Delta(G^\vee)$, this does not prove $\Delta > 0$. The Langlands thread connects G- and $G^\vee$-YM but does not construct the gap. The gap requires dimensional transmutation (Gribov mass $\gamma \to \Lambda_{QCD}$), which is outside the Langlands framework.

3 Riemann and Ramanujan

The Riemann zeta function $\zeta(s) = L(s, \mathbf{1})$ is the automorphic L-function of the trivial representation of $GL(1, \mathbb{A}_\mathbb{Q})$. GRH ($|\mathrm{Re}(\rho) - 1/2| = 0$) is the analogue, for $GL(1)$, of the Ramanujan conjecture for $GL(n)$.

Conjecture 8 (Ramanujan-Petersson)**.** *Let π be a cuspidal automorphic representation of $GL(n, \mathbb{A}_\mathbb{Q})$ with unramified local components $\pi_p = \mathrm{Ind}(\chi_1 \otimes \cdots \otimes \chi_n)$. Then the Satake parameters $\alpha_{i,p}$ satisfy $|\alpha_{i,p}| = 1$ for all p.*

Theorem 9 (Langlands-to-RH reduction)**.** *If the Ramanujan-Petersson conjecture holds for all cuspidal automorphic representations of $GL(n, \mathbb{A}_\mathbb{Q})$ for all n, then GRH holds for all automorphic L-functions. In particular, it implies RH for $\zeta(s)$.*

Proof. For $L(s, \pi)$ with local factors $L_p(s, \pi) = \prod_i (1 - \alpha_{i,p} p^{-s})^{-1}$: Ramanujan gives $|\alpha_{i,p}| = 1$, so $L_p(s, \pi)$ has zeros only on $\mathrm{Re}(s) = 0$ (as a polynomial in p^{-s}). The global L-function, a product of such local factors, then has zeros only on $\mathrm{Re}(s) = 1/2$ by the functional equation $s \leftrightarrow 1 - s$. For $\zeta(s)$ ($n = 1$, trivial representation), this is exactly RH. □

Remark 10 (Status of Ramanujan)*.*

- $n = 1$: trivial (Hecke characters). ✓
- $n = 2$, holomorphic forms weight $k \geq 2$: Deligne (1974). ✓
- $n = 2$, Maass forms: open. The best bound is $|\alpha_{i,p}| \leq p^{7/64}$ (Kim-Sarnak [5]).
- $n \geq 3$: partial results via Langlands functoriality (Kim-Shahidi: Sym^n for $n \leq 4$, symmetric power functoriality).
- General n: open.

GRH for $\zeta(s)$ specifically may be provable without full Ramanujan; the Hilbert-Pólya approach (construct H) is orthogonal to the Langlands approach.

Remark 11 (Why the function field works)*.* Over $\mathbb{F}_q(C)$, Ramanujan is Deligne's theorem (Weil II, 1980): the eigenvalues of Frobenius on ℓ-adic cohomology are algebraic integers all of absolute value $q^{w/2}$ (purity). This gives $|\alpha_{i,p}| = 1$ immediately. Over $\mathbb{Q}$, there is no Frobenius: it is local (at each prime p), not global, and the cohomological interpretation requires the full machinery of motives (largely conjectural).

4 BSD and Arithmetic Langlands

BSD is a statement about the arithmetic Langlands correspondence for $GL(2)$ applied to the L-function $L(E, s)$ of an elliptic curve.

Proposition 12 (BSD as arithmetic Langlands)**.** *Let $E/\mathbb{Q}$ have modular form f_E of weight 2 and conductor N (Wiles-BCDT [6, 7]). Then $L(E, s) = L(f_E, s) = L(\pi_E, s)$ where π_E is the automorphic representation of $GL(2, \mathbb{A}_\mathbb{Q})$ generated by f_E. BSD asserts that the arithmetic rank of E equals the analytic rank $\mathrm{ord}_{s=1} L(\pi_E, s)$.*

The Langlands framework gives:

1. **Rank 0**: $L(\pi_E, 1) \neq 0$ implies $E(\mathbb{Q})$ is finite (Kolyvagin, using GL(2) Euler system machinery).

2. **Rank 1**: Gross-Zagier formula relates $L'(\pi_E, 1)$ to the height of a Heegner point; Kolyvagin bounds the Selmer group.

3. **Rank** $r \geq 2$: GGZ_r is the arithmetic Langlands functoriality statement for $U(r) \times U(r+1) \hookrightarrow U(r+1) \times U(r+1)$ — a period identity for the diagonal embedding of unitary groups (arithmetic GGP).

Remark 13 (Geometric vs. arithmetic Langlands for BSD)*.* The geometric Langlands (Theorem 2) gives the categorical structure of D-modules on $\mathrm{Bun}_{GL(2)}(C)$ for a curve C over $\mathbb{C}$. This is the wrong geometry for BSD: what is needed is the arithmetic of $E(\mathbb{Q})$, which involves the Galois group $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$. Geometric Langlands (over $\mathbb{C}$) has no Galois action; it does not see the arithmetic of rational points.

BSD for rank ≥ 2 is blocked at the arithmetic GGP level: constructing the Euler system for $U(r) \times U(r+1)$ over $\mathbb{Q}$ requires Galois-equivariant cohomology classes that geometric tools (which work over $\mathbb{C}$) do not produce.

5 The Arithmetic Wall

All three connections — Langlands-to-RH (via Ramanujan), Langlands-to-BSD (via arithmetic GGP), and Langlands-to-YM (via Kapustin-Witten) — encounter the same obstacle at the point of contact with the Clay problem.

Theorem 14 (The arithmetic wall)**.** *The geometric Langlands program (proved) does not imply any of the following:*

1. *GRH or RH.*

2. *BSD for any rank r.*

3. *A mass gap for pure Yang-Mills.*

The obstacle in each case is the same: the arithmetic of $\mathbb{Q}$ (Galois actions, Frobenius at primes, global-local compatibility over number fields) is not accessible to geometric tools working over $\mathbb{C}$.

Argument. (1): Ramanujan over $\mathbb{Q}$ requires controlling Frobenius eigenvalues globally. The geometric Langlands gives a derived equivalence of categories over $\mathbb{C}$; it has no $\mathbb{F}_q$ structure and no Frobenius.

(2): BSD requires Galois-equivariant Euler systems over $\mathbb{Q}$. The geometric Langlands equivalence has no Galois group action on either side (both sides are over $\mathbb{C}$, not over $\mathbb{Q}$ or $\overline{\mathbb{Q}}$).

(3): The pure YM connection to geometric Langlands goes through $\mathcal{N} = 4$ SYM (Remark 5). The mass gap requires dimensional transmutation in the non-supersymmetric theory; this is a dynamical statement absent from the exact conformal field theory context of geometric Langlands. □

6 Navier-Stokes: The Outlier

Navier-Stokes has no connection to the Langlands program. The equations are:

$$\partial_t u + (u \cdot \nabla)u - \nu \Delta u + \nabla p = f, \quad \nabla \cdot u = 0.$$

There is no reductive group, no automorphic representation, no L-function, no dual group, no Frobenius. The difficulty is purely analytic: vortex stretching in three dimensions generates the supercritical cubic term that prevents closure of the enstrophy inequality.

Remark 15. The isolation of NS from the Langlands thread is not a failure of abstraction but a genuine structural separation. NS is a nonlinear PDE in real analysis; the other three problems (RH, BSD, YM) involve the arithmetic of $\mathbb{Q}$ or its global structure. The unified language for RH, BSD, and YM is automorphic; for NS it is PDE regularity. These are genuinely different subjects.

7 Classification via Langlands

Problem	Langlands side	Connection	Proved?	Closes?
RH	$GL(1)$ automorphic	Ramanujan	No (Maass)	No
YM	G-bundles / S-duality	Kapustin-Witten	Geom. only	No
BSD	$GL(2)$, unitary	Arithmetic GGP	Rank ≤ 2	No
NS	None	—	—	—

Theorem 16 (Langlands thread summary)**.** *The three Clay problems with a Langlands connection (RH, YM, BSD) share the following structure:*

1. *The geometric/categorical Langlands is proved (Theorem 2).*
2. *The connection to each problem runs through arithmetic Langlands, which is open.*
3. *Each problem's bound step (in the adaptive regularization template) is exactly the arithmetic Langlands obstacle:*
   - *RH: Ramanujan for Maass forms (Galois eigenvalues over $\mathbb{Q}$).*
   - *BSD: Arithmetic GGP for $U(r) \times U(r+1)$, $r \geq 3$ (Euler systems over $\mathbb{Q}$).*
   - *YM: Gribov mass persistence (arithmetic of the non-perturbative vacuum over $\mathbb{R}^4$).*
4. *Navier-Stokes has no Langlands structure; its bound step is of a different kind.*

Remark 17 (What proving arithmetic Langlands would give). Full arithmetic Langlands — the global Langlands correspondence over $\mathbb{Q}$ for all reductive G — would imply:

- GRH (via Ramanujan for all automorphic L-functions);
- BSD for all ranks (via arithmetic GGP as a functoriality statement);

- Partial implications for YM (via the arithmetic structure of the gauge vacuum).

Full arithmetic Langlands is believed to be a problem of comparable difficulty to the problems it would solve. It is the correct unified form of the open question.

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